Generative AI-Assisted Malware Disassembly and Remediation

How LLMs accelerate reverse engineering and patching — and where they still fail

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2026-11-08

M.Sc. in Cybersecurity — Malware Analysis

Agenda

1. Context & Motivation
2. The Toolchain
3. Manual vs. AI-Augmented Workflow
4. Evidence: Speed, Accuracy & Hallucinations
5. Live Demo
6. Takeaways

1. **Context & Motivation**
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Reverse engineering is slow, scarce and expensive

- Malware triage and patching are **bottlenecked by human experts**: disassembly, decompilation and dynamic analysis take **hours to days** per sample.
- Volume keeps growing: **hundreds of thousands** of new samples per day; a large **backlog of unpatched CVEs**.
- Skills are rare: reading x86/ARM assembly and noisy decompiler pseudo-code is hard and error-prone.

The question: can generative AI shrink the time-to-understanding and time-to-patch — without lying to us?

The promise of LLMs in this workflow

LLMs can act as a **force multiplier** on top of existing tools:

- **Explain** decompiled pseudo-code in plain language.
- **Rename** functions / variables and **annotate** structures to recover intent.
- **Hypothesize** the exploited bug and **draft** a candidate mitigation or patch.

Framing for this talk: an LLM is an **accelerator and co-pilot**, **not** an autonomous replacement for the analyst. Every output must be **verified against the binary**.

The Toolchain

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The classic RE stack (where we start)

Static analysis

- Disassemblers / decompilers: **Ghidra** (NSA, free), IDA Pro, Binary Ninja.
- Decompiler output = **pseudo-C** reconstructed from machine code: approximate, no original names/comments.
- `strings`, PE/ELF headers, imports.

Dynamic analysis

- Debuggers: x64dbg, GDB.
- Sandboxes: Cuckoo / CAPE, isolated VMs.
- Network / behaviour capture.

Pseudo-code ≠ source code. The decompiler guesses types and control flow; this is exactly the noisy input we ask an LLM to interpret.

The LLM-augmented stack

The LLM is **wired into** the analysis tool, not used by copy-paste:



- **MCP (Model Context Protocol)** bridges, e.g. **GhidraMCP** (LaurieWired), expose Ghidra's API (list/decompile functions, rename, cross-refs) to an LLM agent.
- The agent can **iterate**: decompile → explain → rename → re-read → propose patch.
- Keeps a **human in the loop** who accepts or rejects each change.

Manual vs. AI-Augmented Workflow

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Two lanes, same destination

Manual (traditional)

1. Load binary, map functions by hand.
2. Read assembly / pseudo-code line by line.
3. Guess intent, rename, take notes.
4. Reproduce behaviour in a sandbox.
5. Locate the bug; design a mitigation.

Hours–days; depth depends on expert skill.

AI-augmented

1. Agent auto-decompiles & summarizes functions.
2. LLM explains + renames in bulk.
3. Analyst steers questions (“what triggers this?”).
4. LLM drafts IOCs / YARA / patch.
5. **Analyst verifies** against the binary.

Minutes for triage; verification still required.

Where the AI lane helps — and where it doesn't

Dimension	Manual	AI-augmented
Speed of first-pass triage	Slow	**Fast**
Comprehension for non-experts	Low	**Much higher**
Breadth / coverage of code	Limited by time	**Broad**
Correctness guarantee	**Analyst-verified**	Needs verification
Risk of confident errors	Low	**Hallucinations**

Net effect: AI moves effort from **reading** to **verifying**.

Evidence: Speed, Accuracy & Hallucinations

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Speed & comprehension — what studies show

- **Human–LLM teaming study** (NDSS 2026, 48 participants): novices' comprehension rose by **98%**; functions implementing **known algorithms** were triaged up to **2.4× faster**.
- **But**: experts gained little, and some outputs contained **harmful hallucinations** and misleading suggestions.

Biggest, most reliable win: **lowering the barrier to entry** and speeding up **first-pass triage** — not replacing expert judgement.

Remediation & patching — still hard

- **PatchEval** (2025, 1,000 real CVEs): the best model/agent fixed only **23%** of vulnerabilities. Models are **complementary** — 98 bugs fixed by **at least one** model.
- **Architecture studies** of LLM patching compare fixed-workflow vs. single-/multi-agent vs. general code agents, trading **accuracy vs. time & token cost**.

Takeaway: the LLM **proposes** a patch; a human (and a test-suite) **disposes**.

The hallucination problem

Failure modes reported repeatedly in the literature:

- **Invented logic**: plausible-sounding but wrong explanations.
- **Misleading renaming**: a function named `decrypt_payload` that doesn't.
- **Low recompilability**: DecLLM (ISSTA 2025) raised recompilable decompiled code from **9% → 25%** — i.e. most output still **doesn't even rebuild**.

Mitigation = verify against ground truth: recompile, diff behaviour, symbolic execution, unit tests. This is the **same reason** LLMs remain unreliable for **automated malware classification**.

Live Demo

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5. **Live Demo**
6. Takeaways

Demo — before / after

The "before": in 2017, stopping **WannaCry** meant an analyst reading the malware's code live during a global outbreak to find the **kill-switch domain**. Manual, under pressure, hours of work.

The "after" (our demo, stopwatch on screen):

1. **Act A — Accelerated understanding:** Ghidra + LLM (via GhidraMCP) explains a documented sample in an **isolated VM** → behaviour summary, **IOCs** and a **YARA** rule + mitigation (disable SMBv1 / apply MS17-010).
2. **Act B — End-to-end remediation:** our **own** vulnerable toy binary → LLM spots the bug in the pseudo-code, **proposes the patch**, we recompile → the crash is gone.

Live Demo

▸ Play recorded demo

(Stopwatch overlay: manual estimate vs. AI-augmented time)

Takeaways

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Takeaways

- LLMs **accelerate** reverse engineering and **draft** remediations — biggest wins in **triage** and for **non-experts**.
- They do **not** replace analysts: patch success is **modest** (23%) and **hallucinations** are real.
- The discipline that makes it work: **keep a human in the loop** and **verify every output against the binary**.
- Same lesson explains why AI still **struggles with reliable malware classification**.

Credits & contributions

Member	Contribution
Author 1	Context & motivation, toolchain, slide design (slides 1–8).
Author 2	Workflow comparison, evidence & hallucination research (slides 9–15).
Author 3	Demo design, recording, remediation & takeaways (slides 16–18).

Tools: Typst + Touying (Metropolis). Sources cited in the speaker notes / script.

Thank you!

Questions?